Functional Form
&
Transformation

Regression

Functional Form & Transformations

Part a) Dealing with numerical variables

* Non-linear relationships (squared, inverse etc)

* Logarithms

Regression

Functional Form & Transformations

Part a) Dealing with numerical variables

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Part b) Dealing with categorical X variables

* Dummy variables
* Interaction variables

Regression

Functional Form & Transformations

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Part b) Dealing with categorical X variables

* Dummy variables

* Interaction variables

Part c) Dealing with categorical Y variables

* Logit models

Regression

Dataset:

Jaybob's Used Car Sales (jaybob.csv)

Variables:

"Price" - advertised sale price ($AUD)

"Age" - model age (yrs)

"Odometer" - odometer reading ('000 kms)

"Pink slip" - presence of RWC (1= yes, 0=no)

"Sold" - whether car sold (1=yes, 0=no)

Regression

Dataset:

Jaybob's Used Car Sales (jaybob.csv)

	A	B	C	D	E	F
1	Car ID	Price	Age	Odometer	Pink slip	Sold?
2	1	$ 1,000	28	30.298	1	1
3	2	$ 9,000	40	19.647	1	0
4	3	$ 500	58	170.270	0	1
5	4	$ 3,000	12	68.394	1	1
6	5	$ 9,500	3	11.662	0	0
7	6	$ 1,500	23	87.973	0	0
8	7	$ 4,000	4	3.496	1	0
9	8	$ 2,000	13	40.986	1	1
10	9	$ 2,500	5	21.098	1	1

Numerical variables

Model 1

$$Price_i = \beta_0 + \beta_1 Age_i + \beta_2 Odometer_i + \varepsilon_i$$

Numerical variables

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DV: Price	*Coef*	*SE*	*t*	*P-value*
Intercept	4615.901	792.153	5.83	0.0000
Age	98.922	29.974	3.30	0.0014
Odometer	-23.029	6.284	-3.66	0.0004

$R^2 = 0.171$

Numerical variables

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$$\widehat{Price}_i = 4615.9 + 98.9(Age_i) - 23.0(Odometer_i)$$

Numerical variables

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$$\widehat{Price}_i = 4615.9 + 98.9(Age_i) - 23.0(Odometer_i)$$

For every additional year in age, the car can be expected to increase in price by $98.92, on average, holding odometer constant.

Numerical variables

Model 1

$$Price_i = \beta_0 + \beta_1 Age_i + \beta_2 Odometer_i + \varepsilon_i$$

DV: Price	Coef	SE	t	P-value
Intercept	4615.901	792.153	5.83	0.0000
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$$R^2 = 0.171$$

$$\widehat{Price}_i = 4615.9 + 98.9(Age_i) - 23.0(Odometer_i)$$

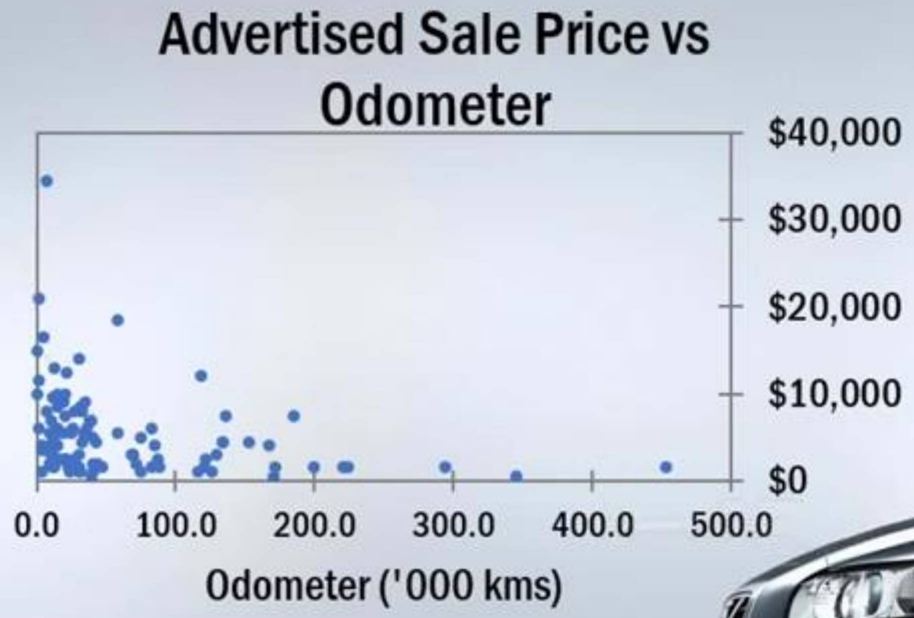
For every additional thousand km on the odometer, the price is expected to decrease by $23.03, on average, holding age constant.

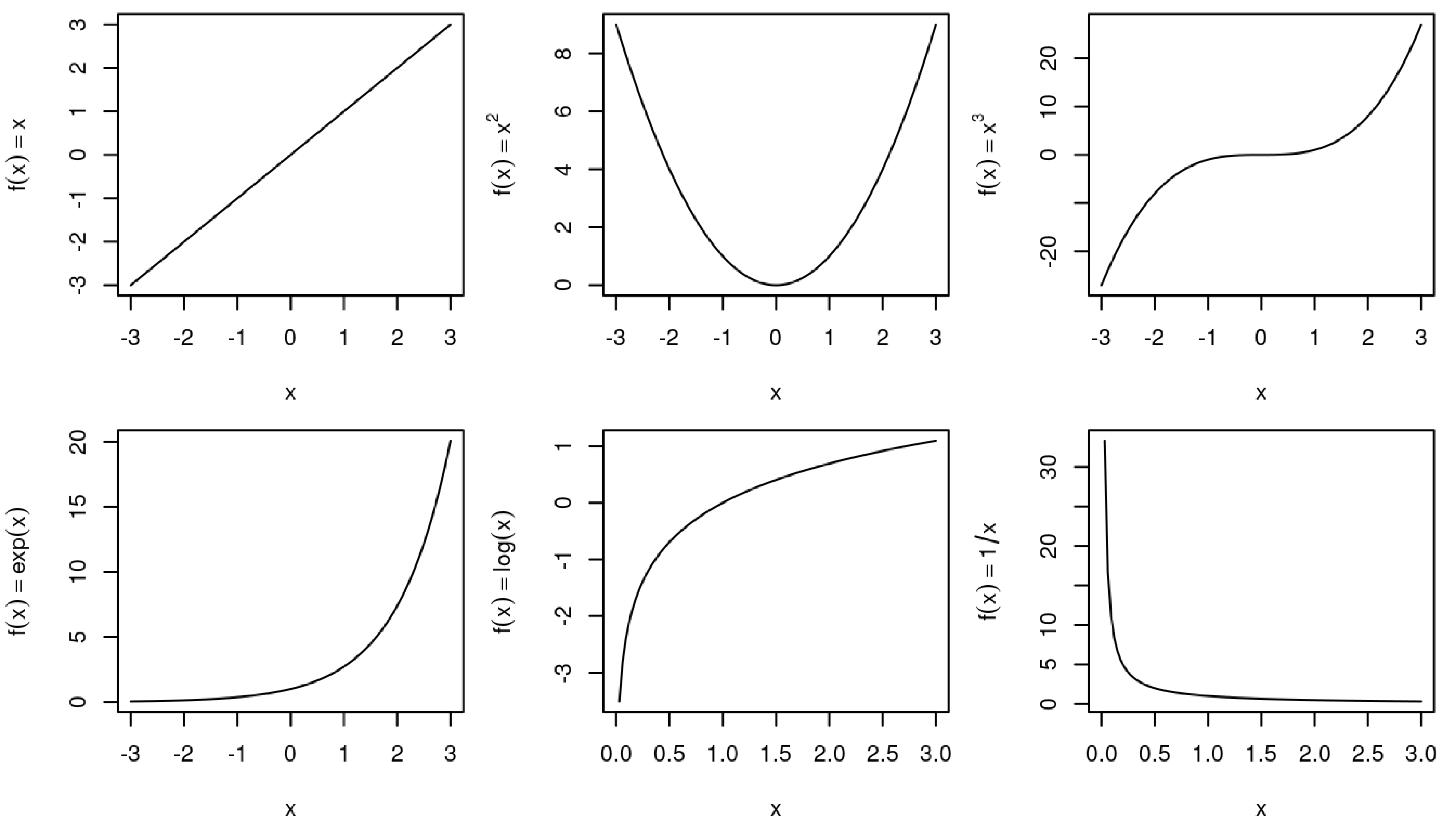
Numerical variables

<u>Check scatter plots!</u>

Numerical variables

Check scatter plots!

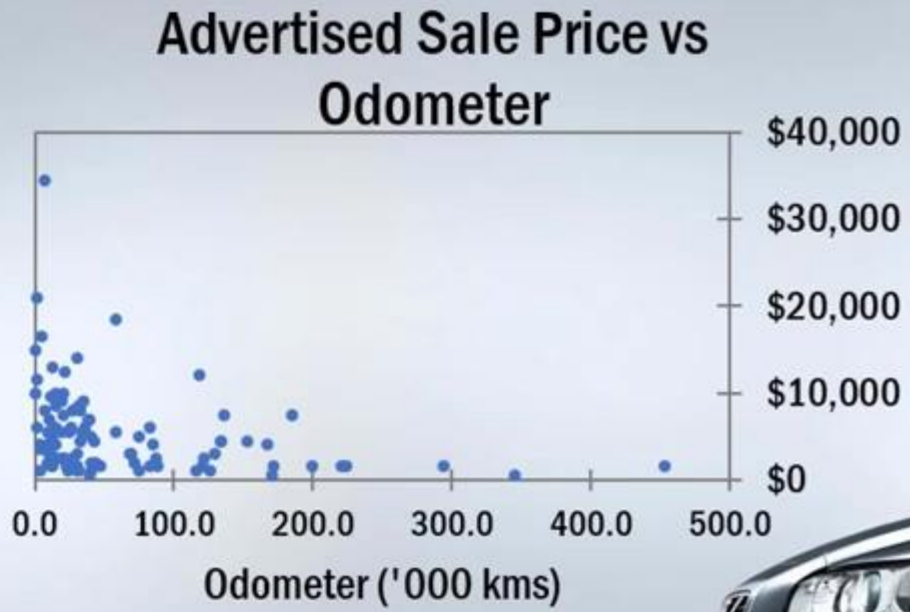




f(x) = x
f(x) = x^2
f(x) = x^3
f(x) = exp(x)
f(x) = log(x)
f(x) = 1/x
x

Numerical variables

Check scatter plots!



$$Price_i = \beta_0 + \beta_1 Age_i + \beta_2 Age_i^2 + \beta_3 \left(\frac{1}{Odometer_i}\right) + \varepsilon_i$$

Numerical variables

Model 2

$$Price_i = \beta_0 + \beta_1(Age_i) + \beta_2(Age_i)^2 + \beta_2\left(\frac{1}{Odometer_i}\right) + \varepsilon_i$$

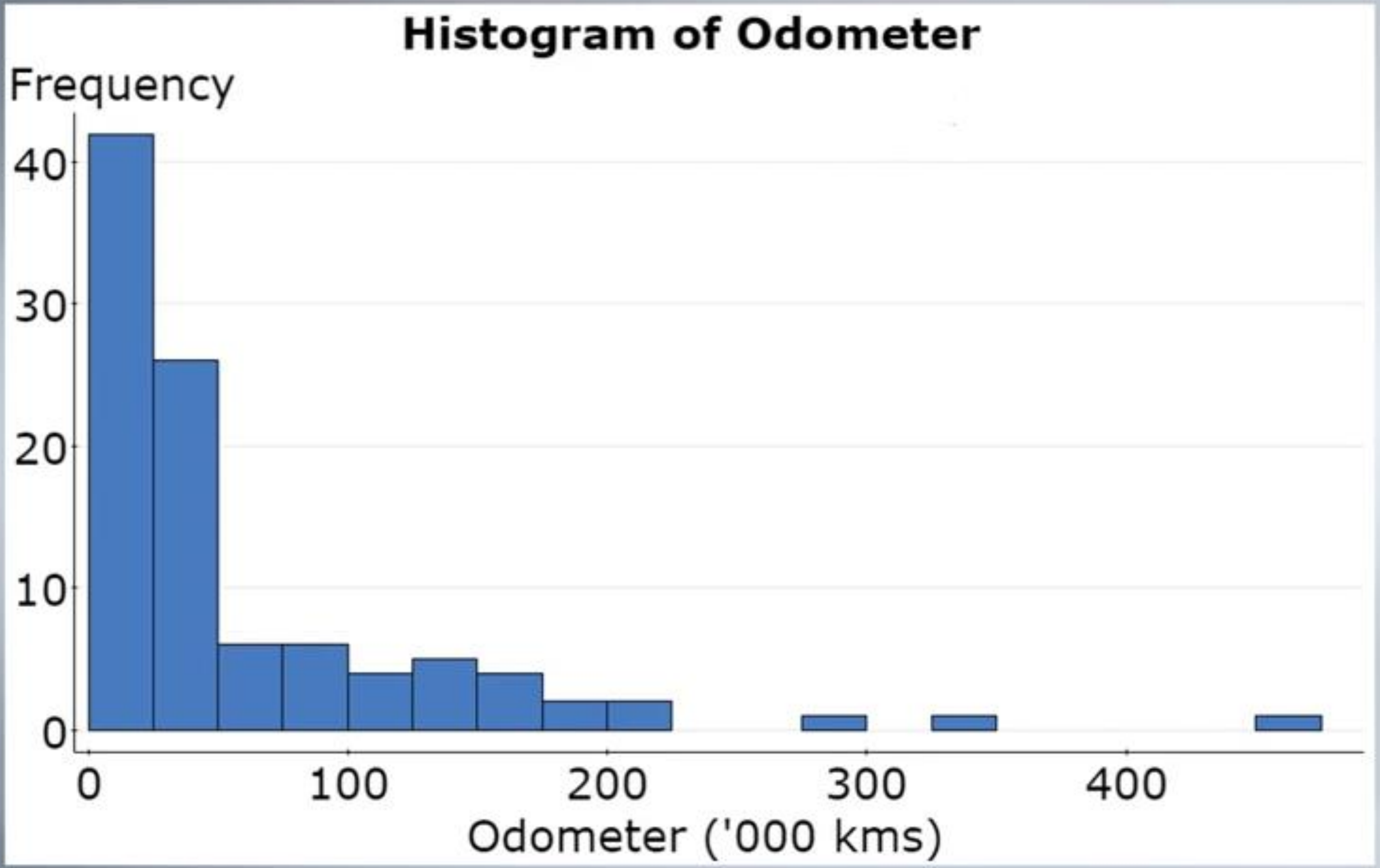
DV: Price	*Coef*	*SE*	*t*	*P-value*
Intercept	8809.879	806.803	10.92	0.0000
Age	-429.664	56.393	-7.62	0.0000
Age2	7.318	0.735	9.96	0.0000
1/Odometer	1942.243	676.217	2.87	0.0050

$$\mathrm{R}^2 = 0.585$$

$$\widehat{Price}_i = 8809.9 - 429.7(Age_i) + 7.3(Age_i)^2 + 1942.2\left({}^{1}/_{Odometer_i}\right)$$

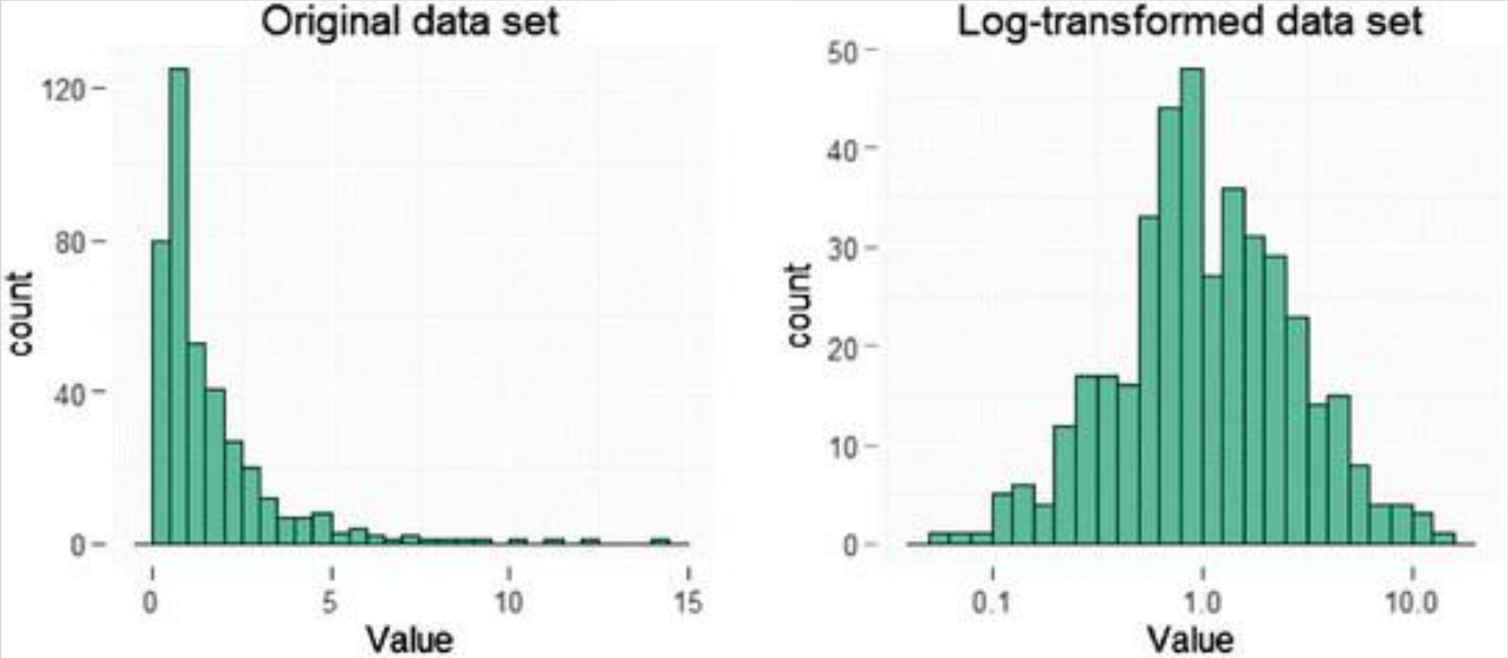
Numerical variables

Logarithms



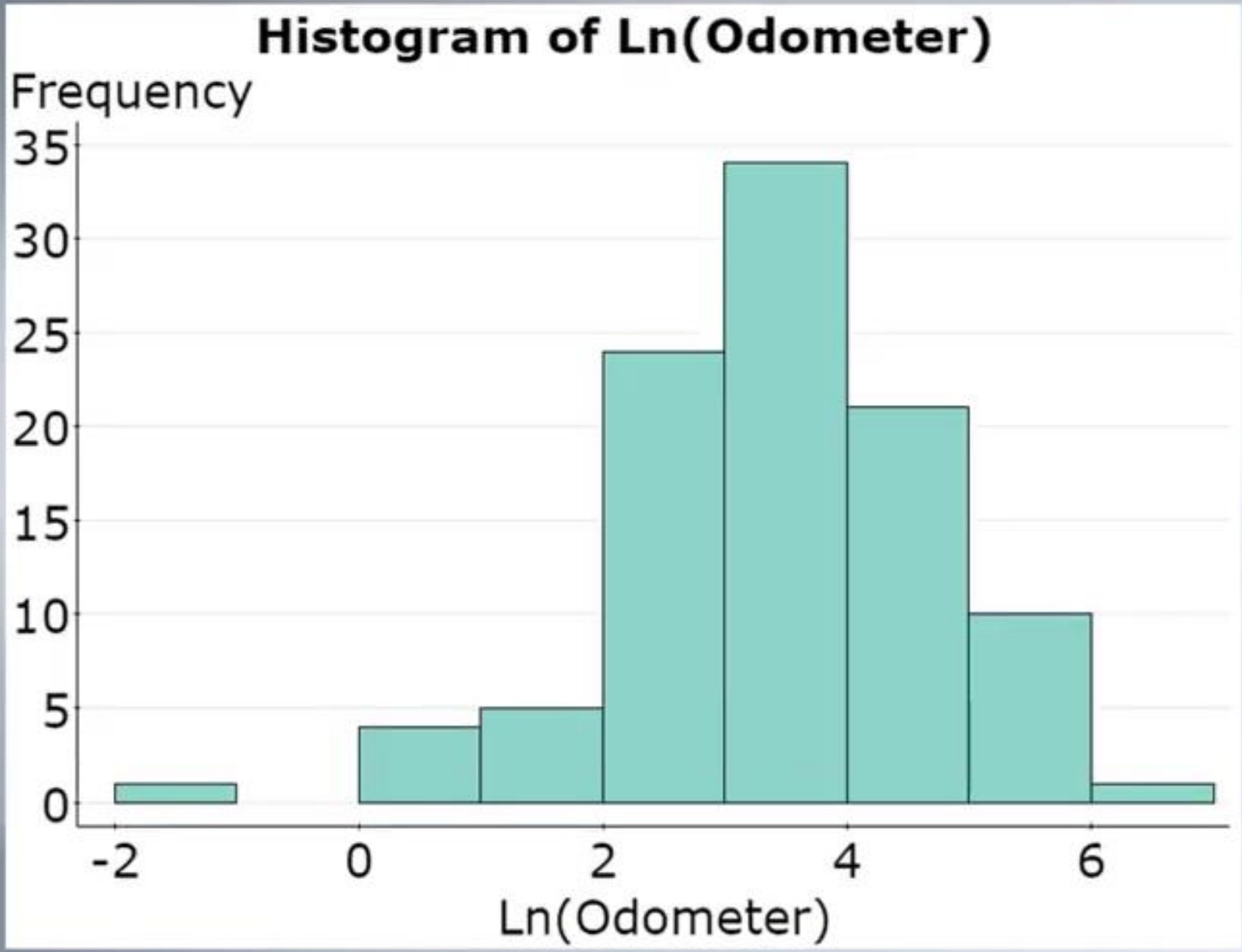
- Used mostly because of skewed distribution. Logarithm naturally reduces the dynamic range of a variable so that the differences are preserved while the scale is not that dramatically skewed.

- Imagine some people got 100,000,000 loan and some got 10000 and some 0. Any feature scaling will probably put 0 and 10000 so close to each other as the biggest number anyway pushes the boundary. Logarithm solves the issue.



Numerical variables

Logarithms



Numerical variables

Model 3

$$Price_i = \beta_0 + \beta_1(Age_i) + \beta_2(Age_i)^2 + \beta_3 Ln(Odometer_i) + \varepsilon_i$$

Numerical variables

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$$Price_i = \beta_0 + \beta_1(Age_i) + \beta_2(Age_i)^2 + \beta_3 Ln(Odometer_i) + \varepsilon_i$$

DV: Price	*Coef*	*SE*	*t*	*P-value*
Intercept	11863.069	940.941	12.61	0.0000
Age	-365.576	58.864	-6.21	0.0000
Age2	6.628	0.749	8.85	0.0000
Ln(Odometer)	-1079.375	272.050	-3.97	0.0001

$$R^2 = 0.613$$

$$\widehat{Price}_i = 11863.1 - 365.6\,(Age_i) + 6.63\,(Age_i)^2 - 1079.4\,Ln(Odometer_i)$$

Numerical variables

Model 3

$$\widehat{Price}_i = 11863.1 - 365.6\,(Age_i) + 6.63\,(Age_i)^2$$
$$-1079.4\,Ln(Odometer_i)$$

Numerical variables

Model 3

$$\widehat{Price}_i = 11863.1 - 365.6\,(Age_i) + 6.63\,(Age_i)^2 - 1079.4\,Ln(Odometer_i)$$

A 1 unit increase in the natural log of the odometer reading decreases the price by $1079.40, on average, holding age constant

Numerical variables

Model 3

$$\widehat{Price}_i = 11863.1 - 365.6\,(Age_i) + 6.63\,(Age_i)^2 \\ -1079.4\,Ln(Odometer_i)$$

A 1% increase in the odometer reading decreases the price by:

1079.4/100 = $10.79

...on average, holding age constant.

Part (b) - Categorical X variables

Categorical X variables

<u>Binary variables</u>

Pink Slip = 1   if car has roadworthy certificate
= 0   otherwise

Categorical X variables

<u>Binary variables</u>

Pink Slip = 1 if car has roadworthy certificate

= 0 otherwise

$$Price_i = \beta_0 + \beta_1(Pink\ Slip_i) + \varepsilon_i$$

Categorical X variables

Binary variables

Pink Slip = 1 if car has roadworthy certificate
= 0 otherwise

$$Price_i = \beta_0 + \beta_1(Pink\ Slip_i) + \varepsilon_i$$

DV: Price	Coef	SE	t	P-value
Intercept	3978.3	1056.29	3.76625	0.00028
Pink slip	1625.6	1203.76	1.35047	0.17998

$$\widehat{Price}_i = 3978 + 1626\ (Pink\ Slip_i)$$

Categorical X variables

<u>Binary variables</u>

Pink Slip = 1 if car has roadworthy certificate

= 0 otherwise

$$\widehat{Price}_i = 3978 + 1626\,(Pink\ Slip_i)$$

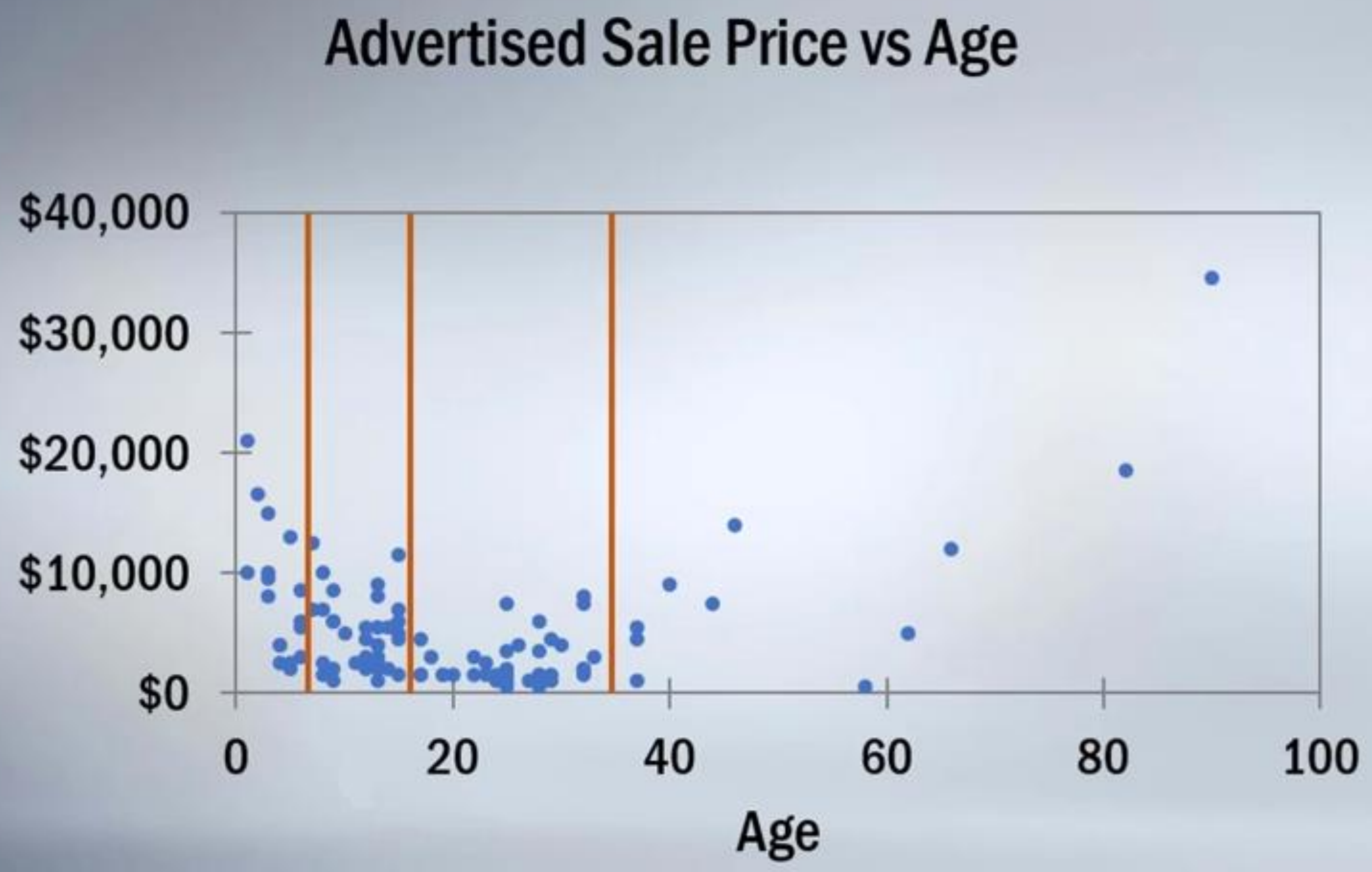
A car with a pink slip would command a sale price $1,626 more than a car without a pink slip, on average.

Categorical X variables

Multi-level categorical variables

Categorical X variables

Multi-level categorical variables



Categorical X variables

Multi-level categorical variables

AgeCat	=	1	if	age <= 5
	=	2	if	5 < age <= 15
	=	3	if	15 < age <= 35
	=	4	if	age > 35

Categorical X variables

Multi-level categorical variables

AgeCat = 1 if age <= 5
= 2 if 5 < age <= 15
= 3 if 15 < age <= 35
= 4 if age > 35

$$Price_i = \beta_0 + \beta_1(Age_i) + \beta_2(Age_i)^2$$
$$+ \beta_3 Ln(Odometer_i) + \beta_4(Pink\ Slip_i) + \varepsilon_i$$

Categorical X variables

Multi-level categorical variables

AgeCat1 = 1 if age <= 5
= 0 otherwise

AgeCat2 = 1 if 5 < age <= 15
= 0 otherwise

AgeCat3 = 1 if 15 < age <= 35
= 0 otherwise

AgeCat4 = 1 if age > 35
= 0 otherwise

$$Price_i = \beta_0 + \beta_1(AgeCat1_i) + \beta_2(AgeCat2_i) + \beta_3(AgeCat3_i) + \beta_4(AgeCat4_i) + \beta_5 Ln(Odometer_i) + \beta_6(Pink\ Slip_i) + \varepsilon_i$$

Categorical X variables

Multi-level categorical variables

$$
\begin{aligned}
Price_i = \beta_0 &+ \beta_1(AgeCat1_i) + \beta_2(AgeCat2_i) \\
&+ \beta_3(AgeCat3_i) + \beta_4(AgeCat4_i) \\
&+ \beta_5 Ln(Odometer_i) + \beta_6(Pink\ Slip_i) + \varepsilon_i
\end{aligned}
$$

BUT!

$$AgeCat1_i = 1 - AgeCat2_i - AgeCat3_i - AgeCat4_i$$

Categorical X variables

Multi-level categorical variables

$$Price_i = \beta_0 + \beta_1(AgeCat1_i) + \beta_2(AgeCat2_i) + \beta_3(AgeCat3_i) + \beta_4(AgeCat4_i) + \beta_5 Ln(Odometer_i) + \beta_6(Pink\ Slip_i) + \varepsilon_i$$

BUT!

$$AgeCat1_i = 1 - AgeCat2_i - AgeCat3_i - AgeCat4_i$$

Dummy variable TRAP

Categorical X variables

Model 6

$$Price_i = \beta_0 + \beta_1(AgeCat2_i) + \beta_2(AgeCat3_i) + \beta_3(AgeCat4_i) + \beta_4 Ln(Odometer_i) + \beta_5(Pink\ Slip_i) + \varepsilon_i$$

Interaction terms

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Intuition

Build a model to explain the salary of all of Google's employees

$$Salary_i = \beta_0 + \beta_1(Employee\ Age_i) + \beta_2(Uni\ degree_i) + \varepsilon_i$$

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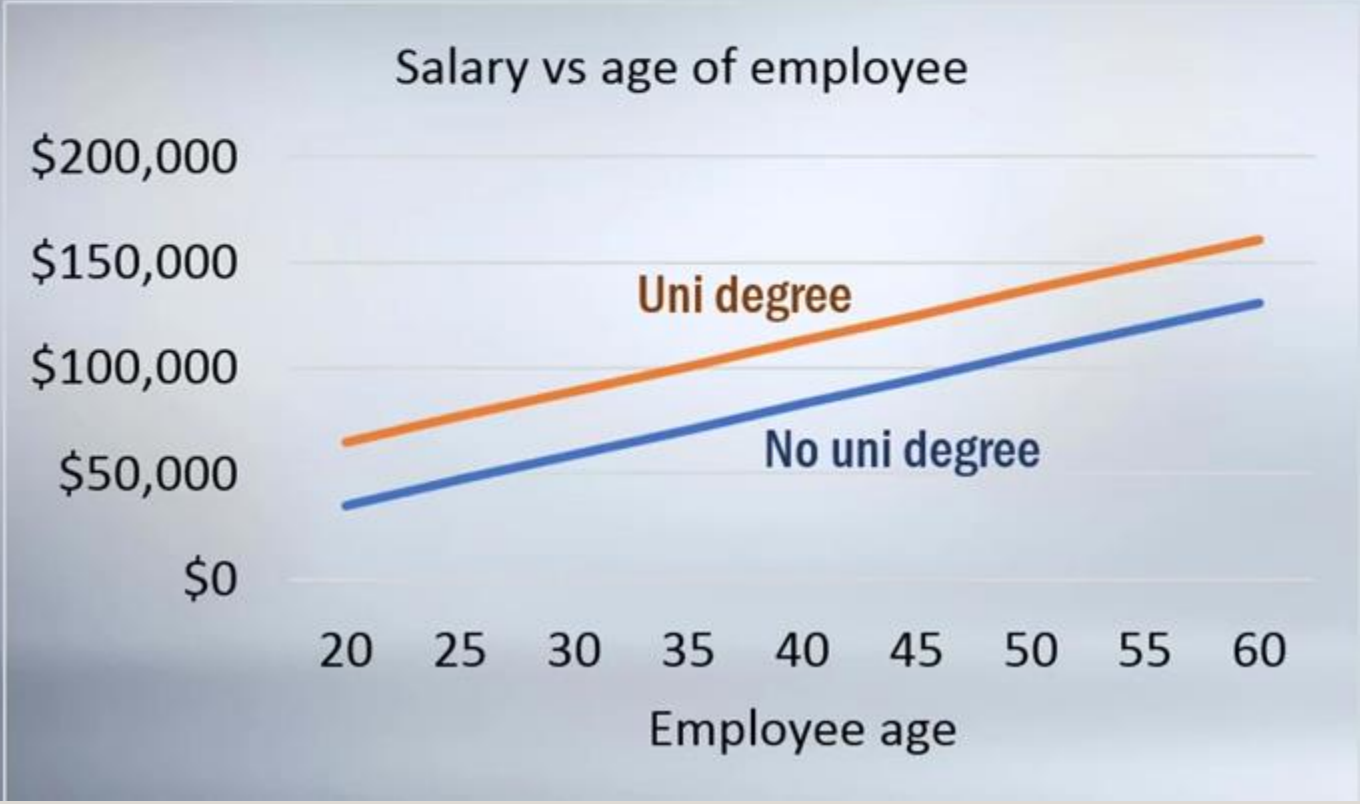

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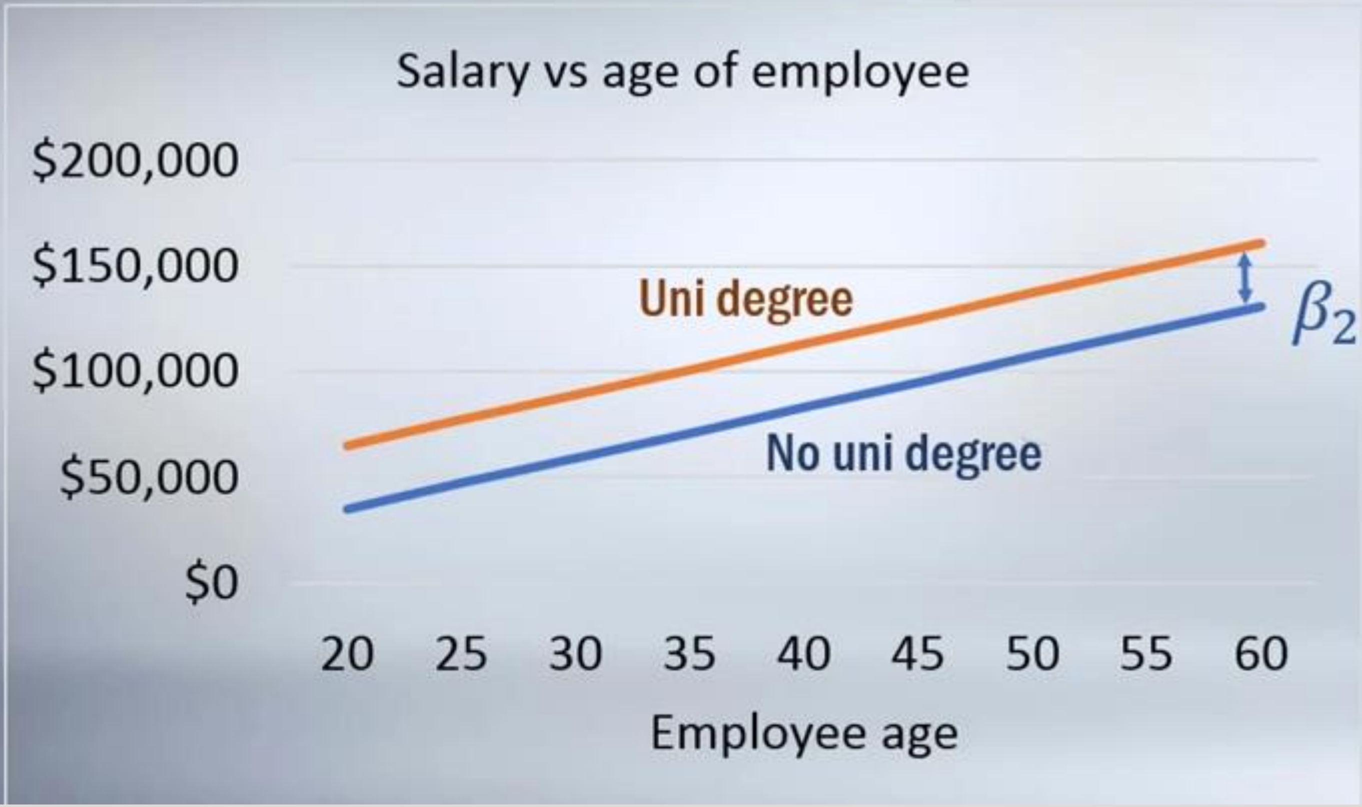

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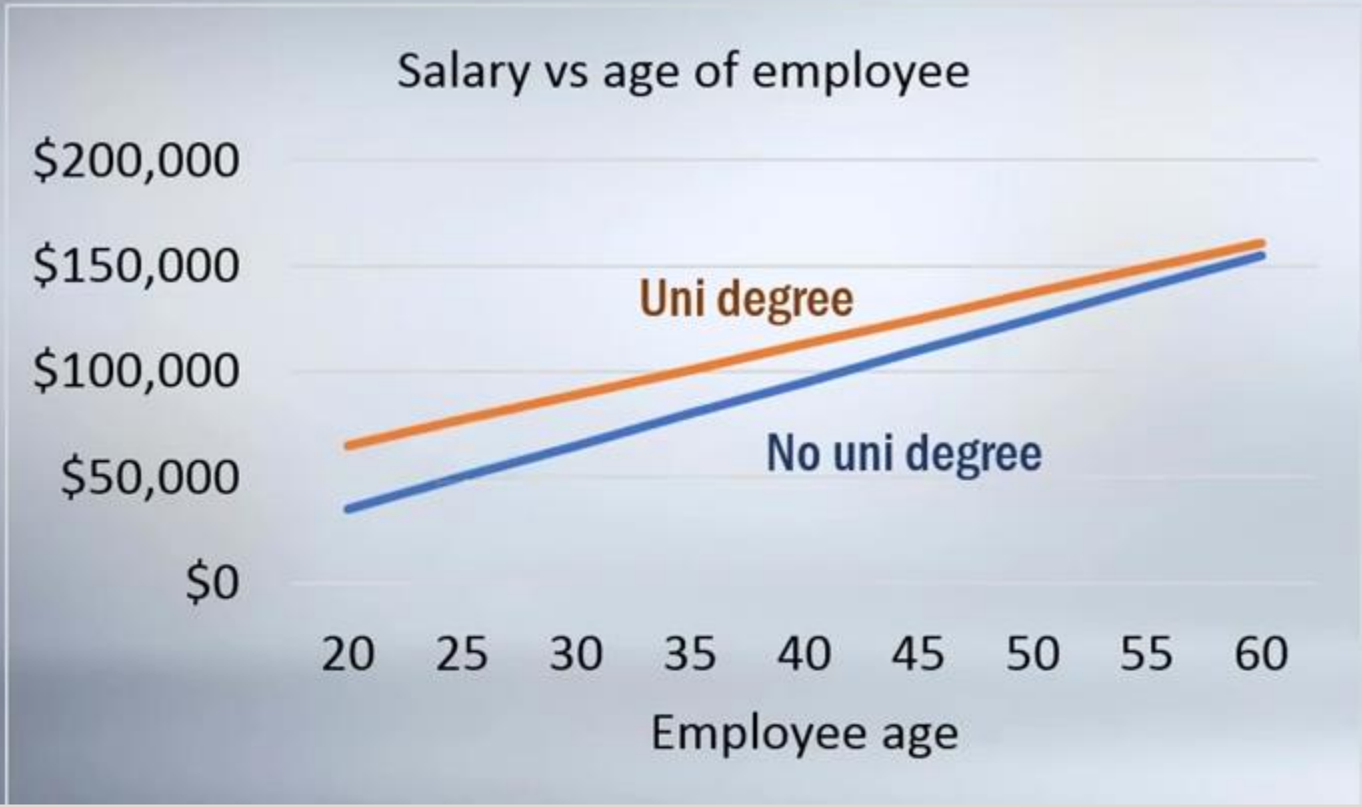
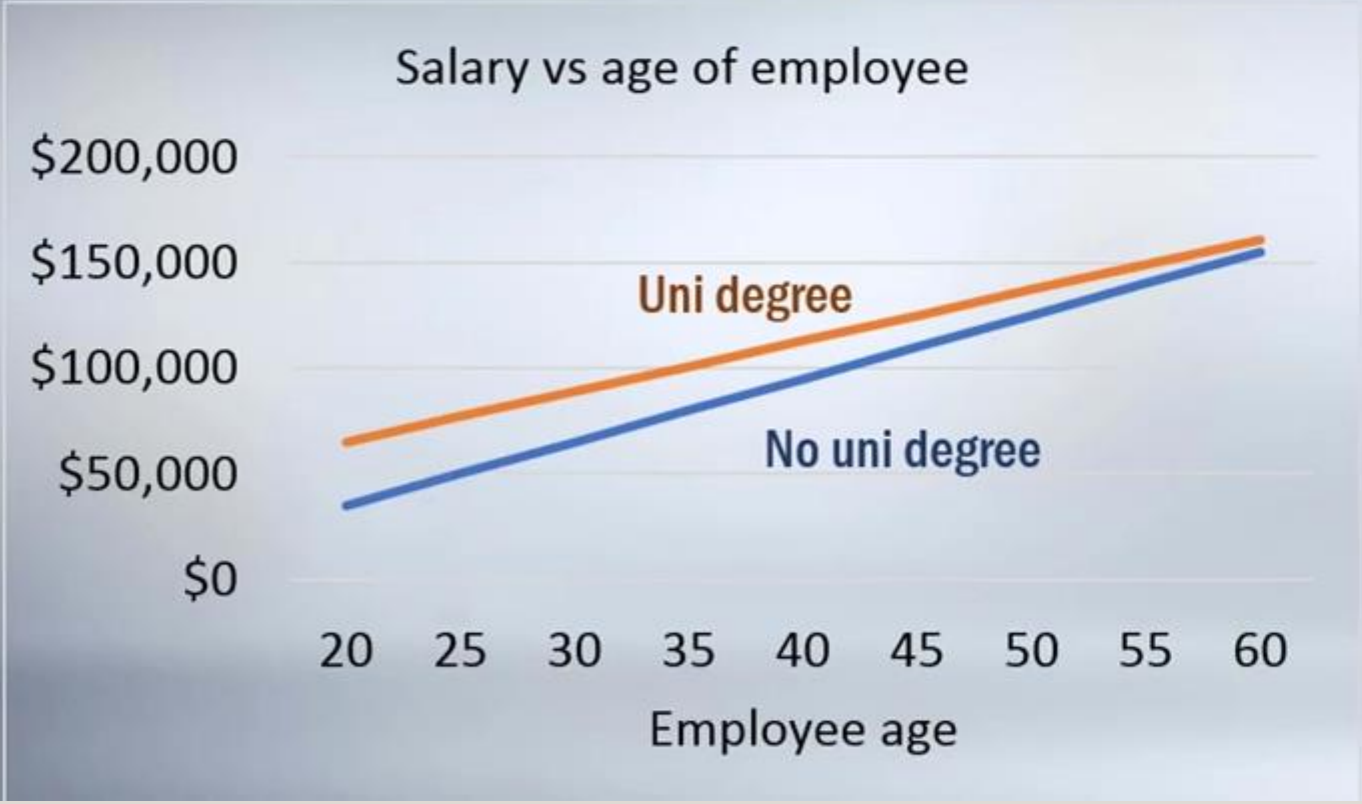


Interaction terms

Intuition

Build a model to explain the salary of all of Google's employees

$$Salary_i = \beta_0 + \beta_1(Employee\ Age_i) + \beta_2(Uni\ degree_i) + \beta_3(Employee\ Age_i) \times (Uni\ degree_i) + \varepsilon_i$$



Interaction terms

<u>Required when:</u>

X1 affects the relationship between X2 and Y

(eg. "Age of employee" affects the relationship between "Attainment of university degree" and "Salary")

<u>Common misconception</u>

"An interaction term is required when X1 and X2 are correlated"

Interaction terms

Model 7

$$Price_i = \beta_0 + \beta_1(AgeCat2_i) + \beta_2(AgeCat3_i) + \beta_3(AgeCat4_i) + \beta_4 Ln(Odometer) + \beta_5(Pink\ Slip_i) + \beta_6(Pink\ Slip_i) \times (AgeCat4_i)$$

DV: Ln(Price)	Coef	SE	t	P-value
Intercept	9.125	0.274	33.28	0.0000
AgeCat2	-0.181	0.238	-0.76	0.4495
AgeCat3	-0.800	0.252	-3.18	0.0020
AgeCat4	-0.390	0.424	-0.92	0.3595
Ln(Odometer)	-0.209	0.059	-3.53	0.0007
Pink slip	0.123	0.182	0.68	0.5005
Pink slip X AgeCat4	1.371	0.453	3.02	0.0032

Non significant?

THANK YOU!